Mock 02

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1 Instructions

Time Allowed 120 Minutes

Total Marks 100

Section A 20 Multiple Choice Questions (2 marks each)

Section B 10 Open Problems (6 marks each)

Note This paper focuses on rigorous proof and synthesis of concepts.

2 Section A: Multiple Choice

1. Logical contrapositive of "Convergent implies Bounded":
   - A. Bounded implies Convergent
   - B. Not Bounded implies Not Convergent
   - C. Not Convergent implies Not Bounded
   - D. Convergent iff Bounded
2. Set Identity (Always true):
   - A. $A \setminus B = A \cap B^c$
   - B. $(A \cup B)^c = A^c \cup B^c$
   - C. $A \subseteq B \iff A \cup B = A$
   - D. $|A \cup B| = |A| + |B|$
3. Cardinality $> \aleph_0$:
   - A. $\mathbb{Q}$
   - B. $\mathbb{Z} \times \mathbb{Z}$
   - C. Finite bit strings
   - D. Functions $f : \mathbb{N} \to \{0, 1\}$
4. The relation $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$ fails:
   - A. Reflexivity
   - B. Symmetry
   - C. Transitivity
   - D. None (It is an Equivalence Relation)
5. If $g \circ f$ is injective:
   - A. g is injective
   - B. f is injective
   - C. Both are injective
   - D. g is surjective
6. No solution exists for:
   - A. $2x \equiv 1 \pmod 5$

- B. $3x \equiv 1 \pmod 6$
- C. $x^2 \equiv 1 \pmod 8$
- D. $x^2 \equiv 2 \pmod 7$

7. Last digit of 7^{2025}:
   - A. 1
   - B. 3
   - C. 7
   - D. 9
8. 10 indistinguishable items into 3 distinguishable boxes (none empty):
   - A. $C(10, 3)$
   - B. $C(9, 2)$
   - C. $C(12, 2)$
   - D. 3^{10}
9. Geometry of roots $z^4 = 1$:
   - A. Rectangle
   - B. Rhombus
   - C. Square in unit circle
   - D. Line
10. Locus $|z - 2| = |z + 2|$:
    - A. Circle
    - B. Real axis
    - C. Imaginary axis
    - D. Ellipse
11. $P(x)$ has integer coefficients, $P(0)$ is odd, $P(1)$ is odd. Integer roots?
    - A. At least one
    - B. None
    - C. Only even
    - D. Only odd

12. Limit of $n!/n^n$:
    - A. 1
    - B. 0
    - C. e
    - D. ∞
13. Limit of $(\sqrt{n^2+n}-n)$:
    - A. 0
    - B. 1
    - C. $1/2$
    - D. ∞
14. Supremum of $\{\frac{n}{n+1}\}$:
    - A. 1
    - B. $1/2$
    - C. 0
    - D. None
15. NOT a subspace of $\mathbb{R}^2$:
    - A. $y = 2x$
    - B. $x + y = 0$
    - C. $x + y = 1$
    - D. $\{(0,0)\}$
16. $\mathbf{a} \cdot \mathbf{b}$ if $|\mathbf{a}| = 3, |\mathbf{b}| = 4$, angle $= 60°$:
    - A. 12
    - B. 6
    - C. $6\sqrt{3}$
    - D. 0
17. "There is a smallest positive real":
    - A. $\exists x \forall y (x \leq y)$
    - B. $\forall x \exists y (y < x)$

- C. $\exists x \forall y (y \leq x)$
- D. $\forall x \exists y (x < y)$

18. Uncountable set A minus Countable set B is:
    - A. Finite
    - B. Countable
    - C. Uncountable
    - D. Empty
19. Strong Induction step proves:
    - A. $P(k) \implies P(k+1)$
    - B. $(P(1) \wedge \cdots \wedge P(k)) \implies P(k+1)$
    - C. $P(1) \implies P(n)$
    - D. $P(k) \implies P(k+2)$
20. Zero divisor in $\mathbb{Z}_{12}$:
    - A. 1
    - B. 5
    - C. 7
    - D. 4

3 Section B: Open Problems

3.1 Problem B1 (Induction)

Prove

$$2^n < n!$$

for

$$n \geq 4$$

.

3.2 Problem B2 (Set Theory)

Prove

$$f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$$

.

3.3 Problem B3 (Number Theory)

Prove there are no integer solutions for

$$15x^2 - 7y^2 = 9$$

.

3.4 Problem B4 (Combinatorics)

Solve

$$x_1 + x_2 + x_3 + x_4 = 20$$

with constraints:

$$x_1 \geq 2, \quad x_2 \geq 0, \quad x_3 \geq 1, \quad x_4 \geq 3$$

3.5 Problem B5 (Complex Geometry)

Identify and sketch the locus defined by:

$$|z - 2i| = 2|z + i|$$

3.6 Problem B6 (Polynomials)

Given

$$P(x) = x^3 + px + q$$

, find

$$\sum x_i^2$$

(the sum of the squares of the roots) in terms of p.

3.7 Problem B7 (Sequences)

Prove that the sequence defined by

$$a_n = \sqrt{2}, \quad a_{n+1} = \sqrt{2 + a_n}, \quad n \in \mathbb{N}$$

converges to 2.

3.8 Problem B8 (Limits)

Prove

$$\lim_{x \to 2} x^2 = 4$$

using the $\varepsilon - \delta$ definition.

3.9 Problem B9 (Vectors)

Decompose the vector $(1, 2, 3)$ into components parallel and perpendicular to $(1, 0, -1)$.

3.10 Problem B10 (Cardinality)

Prove

$$|(0, 1)| = |[0, 1]|$$

using the Cantor-Schröder-Bernstein (CSB) Theorem.

4 Answers

4.1 Part 1: Section A (MCQ) Answers

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	6	B	11	B	16	B
2	A	7	C	12	B	17	A
3	D	8	B	13	C	18	C
4	D	9	C	14	A	19	B
5	B	10	C	15	C	20	D

4.1.1 Explanatory Notes

- **Q6:** $3x \equiv 1 \pmod 6$ implies $3x = 6k + 1 \implies 3(x - 2k) = 1$. This is impossible (a multiple of 3 cannot equal 1).
- **Q11:** If $P(0)$ is odd, the constant term is odd. If $P(1)$ is odd, the sum of coefficients is odd. If an integer root k exists, $P(k) = 0$ (which is even). Parity arguments show this is impossible.
- **Q20:** $4 \times 3 = 12 \equiv 0 \pmod{12}$, therefore 4 is a zero divisor.

4.2 Part 2: Section B Solutions & Rubric

4.2.1 B1: Inequality Induction

- ☐ **(1 pt) Base Case:** $n = 4$. $2^4 = 16$ and $4! = 24$. Since $16 < 24$, the base case holds.
- ☐ **(1 pt) Hypothesis:** Assume $2^k < k!$ for some $k \geq 4$.
- ☐ **(3 pts) Inductive Step:**

$$2^{k+1} = 2 \cdot 2^k < 2 \cdot k!$$

  We need to show $2 \cdot k! < (k+1)!$. Note that $(k+1)! = (k+1)k!$. Since $k \geq 4$, it is clear that $2 < k + 1$. Thus: $2 \cdot k! < (k + 1)k! = (k + 1)!$.
- ☐ **(1 pt) Conclusion:** By PMI, true for all $n \geq 4$.

4.2.2 B2: Set Theory Proof

- ☐ **(3 pts) ($\subseteq$) Direction:** Let $x \in f^{-1}(A \cap B)$. By definition, $f(x) \in A \cap B$. $\implies f(x) \in A \wedge f(x) \in B \implies x \in f^{-1}(A) \wedge x \in f^{-1}(B) \implies x \in f^{-1}(A) \cap f^{-1}(B)$.

- ☐ **(3 pts) ($\supseteq$) Direction:** Let $x \in f^{-1}(A) \cap f^{-1}(B)$. $\implies x \in f^{-1}(A) \wedge x \in f^{-1}(B) \implies f(x) \in A \wedge f(x) \in B \implies f(x) \in A \cap B \implies x \in f^{-1}(A \cap B)$.

4.2.3 B3: Diophantine Impossibility

- ☐ **(2 pts) Analyze Modulo 3:** $15x^2 - 7y^2 = 9 \implies 0 - 7y^2 \equiv 0 \pmod 3$. $-y^2 \equiv 0 \pmod 3 \implies y \equiv 0 \pmod 3$. Let $y = 3k$.
- ☐ **(2 pts) Substitute and simplify:** $15x^2 - 7(9k^2) = 9 \implies 15x^2 - 63k^2 = 9$. Divide by 3: $5x^2 - 21k^2 = 3$.
- ☐ **(2 pts) Analyze Modulo 5:** $0 - 21k^2 \equiv 3 \pmod 5$. $-k^2 \equiv 3 \pmod 5 \implies k^2 \equiv -3 \equiv 2 \pmod 5$. **Check quadratic residues mod 5:** $0^2 = 0, 1^2 = 1, 2^2 = 4, 3^2 = 4, 4^2 = 1$. There is no integer whose square is $2 \pmod 5$. Thus, no solutions exist.

4.2.4 B4: Advanced Combinatorics

- ☐ **(3 pts) Variable Transformation:** Let $y_1 = x_1 - 2, y_2 = x_2, y_3 = x_3 - 1, y_4 = x_4 - 3$, where $y_i \geq 0$. New equation: $(y_1 + 2) + y_2 + (y_3 + 1) + (y_4 + 3) = 20$. $y_1 + y_2 + y_3 + y_4 = 20 - 6 = 14$.
- ☐ **(3 pts) Stars and Bars:** $n = 14, k = 4$. $\binom{n+k-1}{k-1} = \binom{14+3}{3} = \binom{17}{3}$. Calculation: $\frac{17 \cdot 16 \cdot 15}{6} = 17 \cdot 8 \cdot 5 = 680$.

4.2.5 B5: Complex Locus

- ☐ **(2 pts) Algebraic Setup:** $|z - 2i|^2 = 4|z + i|^2$. Let $z = x + iy$. $x^2 + (y - 2)^2 = 4[x^2 + (y + 1)^2]$.
- ☐ **(2 pts) Expansion & Simplification:** $x^2 + y^2 - 4y + 4 = 4(x^2 + y^2 + 2y + 1)$. $x^2 + y^2 - 4y + 4 = 4x^2 + 4y^2 + 8y + 4$. $3x^2 + 3y^2 + 12y = 0 \implies x^2 + y^2 + 4y = 0$.
- ☐ **(2 pts) Result:** Complete square: $x^2 + (y + 2)^2 = 4$. Locus is a circle centered at $(0, -2)$ with radius 2.

4.2.6 B6: Polynomials

- ☐ **(2 pts) Vieta's Formulas:** $\sum x_i = 0$ (no x^2 term) and $\sum x_i x_j = p$.

- ☐ **(2 pts) Square Identity:** $(x_1 + x_2 + x_3)^2 = \sum x_i^2 + 2\sum x_i x_j$. $0^2 = \sum x_i^2 + 2p \implies \sum x_i^2 = -2p$.
- ☐ **(2 pts) Conclusion:** If the sum of squares is 0, then $-2p = 0 \implies p = 0$. (Note: If roots are real, $x_i = 0$; if complex, sum of squares can be zero, but coefficient logic holds).

4.2.7 B7: Sequence Analysis

- ☐ **(2 pts) Boundedness (< 2):** Base $a_1 = \sqrt{2} < 2$. Assume $a_k < 2$. Then $a_{k+1} = \sqrt{2 + a_k} < \sqrt{2+2} = 2$.
- ☐ **(2 pts) Monotonicity (Increasing):** $a_{n+1} > a_n \iff \sqrt{2 + a_n} > a_n \iff 2 + a_n > a_n^2 \iff a_n^2 - a_n - 2 < 0$. Roots of $x^2 - x - 2$ are $2, -1$. Since $0 < a_n < 2$, the parabola is negative, so the inequality holds.
- ☐ **(2 pts) Limit:** Let $L = \lim a_n$. $L = \sqrt{2 + L} \implies L^2 - L - 2 = 0$. $(L-2)(L+1) = 0$. Since $a_n > 0$, $L = 2$.

4.2.8 B8: $\epsilon - \delta$ Proof

- ☐ **(4 pts) Finding δ:** Goal: $|x^2 - 4| < \epsilon \iff |x-2||x+2| < \epsilon$. Restrict $\delta \leq 1 \implies 1 < x < 3 \implies |x+2| < 5$. We need $|x-2| \cdot 5 < \epsilon$. Choose $\delta = \min(1, \epsilon/5)$.
- ☐ **(2 pts) Verification:** Formal write-up of steps.

4.2.9 B9: Vector Decomposition

- ☐ **(3 pts) Parallel Component:** $\mathbf{b} = (1, 0, -1)$, $|\mathbf{b}|^2 = 2$. $\mathbf{a} \cdot \mathbf{b} = 1(1) + 2(0) + 3(-1) = -2$. $\mathbf{a}_{\parallel} = \text{proj}_{\mathbf{b}}\mathbf{a} = \frac{-2}{2}(1, 0, -1) = (-1, 0, 1)$.
- ☐ **(3 pts) Perpendicular Component:** $\mathbf{a}_{\perp} = \mathbf{a} - \mathbf{a}_{\parallel} = (1, 2, 3) - (-1, 0, 1) = (2, 2, 2)$. **Check:** $(2, 2, 2) \cdot (1, 0, -1) = 2 + 0 - 2 = 0$.

4.2.10 B10: Cantor-Schröder-Bernstein

- ☐ **(2 pts) Injection $f : (0, 1) \to [0, 1]$:** Define $f(x) = x$. This is a trivial injection.
- ☐ **(2 pts) Injection $g : [0, 1] \to (0, 1)$:** Map $0 \to 1/2$, $1 \to 1/3$, $1/n \to 1/(n+2)$ for integers $n \geq 2$, identity otherwise. **Alternatively:** $g(x) = \frac{x+1}{3}$ maps $[0, 1] \to [1/3, 2/3] \subset (0, 1)$.

- ☐ **(2 pts) Conclusion:** Since injections exist in both directions, $|(0,1)| = |[0,1]|$ by CSB theorem.